

Multiple Choice (10 marks)

1. You cool a blackbody to half its original temperature. How does its spectrum change?
 - (a) Power emitted is $1/16$ times as high; peak emission wavelength is $1/2$ as long.
 - (b) Power emitted is $1/4$ times as high; peak emission wavelength is 2 times longer.
 - (c) Power emitted is $1/4$ times as high; peak emission wavelength is $1/2$ as long.
 - (d) Power emitted is $1/16$ times as high; peak emission wavelength is 2 times longer.
2. A sand bag has a mass of 5 kg and weight 50 N on Earth. What is the mass and weight of this sand bag on a planet with half the gravity compared to Earth?
 - (a) Mass 5 kg weight 100 N
 - (b) Mass 5 kg weight 50 N
 - (c) Mass 5 kg weight 25 N
 - (d) Mass 10 kg weight 100 N
3. Our current best observations show that Pluto has
 - (a) one medium sized satellite and two small satellites.
 - (b) no satellites.
 - (c) one large satellite and three small satellites.
 - (d) one large satellite.
4. Mars has an atmosphere that is almost entirely carbon dioxide. Why isn't there a strong greenhouse effect keeping the planet warm?
 - (a) Mars does not have enough internal heat to drive the greenhouse effect
 - (b) Mars is too far from the sun for the greenhouse effect to work
 - (c) the greenhouse effect requires an ozone layer which Mars does not have
 - (d) the atmosphere on Mars is too thin to trap a significant amount of heat
5. The famous Drake equation attempts to answer the following question:
 - (a) Will the Sun become a black hole?
 - (b) Is the universe infinitely large?
 - (c) How old is the visible universe?
 - (d) Are we alone in the universe?

True / False (10 marks)

Answer True or False

6. True or False: A black body is an idealized physical object that reflects all electromagnetic radiation, including polarized light.
7. True or False: The Phoenix Mars Lander successfully landed on Mars on May 25, 2008, and was designed to search for ice-rich soil.
8. True or False: A single successful experiment can definitively prove a scientific theory to be correct beyond any doubt.
9. True or False: The Large Magellanic Cloud is a dwarf galaxy that is gravitationally bound to the Milky Way galaxy.
10. True or False: Venus lacks significant seasonal variation because its rotation axis is nearly perpendicular to the plane of the Solar System.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the differences between sidereal and solar days, and critically evaluate the statement: "The time it takes for the Moon to make one circuit of our sky is one solar day."
12. Discuss the reasons behind the differences in composition between the inner and outer planets, focusing on the role of temperature variations in the early solar nebula.

End of Paper